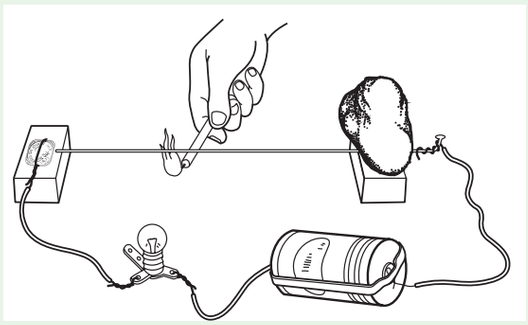
as shown in Figure . The spoke should be parallel to the ground. Place the second wooden block under the free end of the spoke. Wrap some electric wire around the coin (or nail) and place it on the block. You may put a stone over the coin to hold it in place.

Connect a bulb and dry cell to the free ends of the wires connected to the coin and the spoke and make the circuit shown in the figure.



When the tip of the free end of the spoke touches the coin, the circuit is completed and the bulb lights up. Check to ensure this. If the bulb does not light up, it means the circuit is not complete, so check your connections properly. (**Note:** We will learn about electric circuit elaborately in electricity lesson.) Now slide a page of your book between the coin and spoke and then slide it out. That way you would get a gap between the coin and spoke equal to the thickness of the sheet of paper.

❖ Does the bulb light up? If it does not, what could be the reason?

_____

_____

You saw that the bulb does not light up when the spoke does not touch the coin. Now light the candle and heat the spoke with it.

❖ Did the bulb light up after the spoke was heated for some time?

_____

❖ If it did, then explain how the spoke touched the coin after it was heated.

_____

_____

❖ Why does the bulb go off some time after the candle is taken away from the spoke?

_____

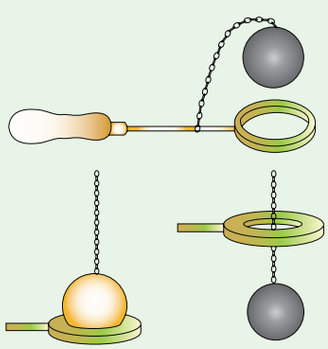
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❖ What happens to the length of the spoke when it is heated or cooled?

_____

Activity 6: Cubical Expansion

Take a metal ring and metal ball of such size that the ball just passes through the ring.



❖ Heat the ball and check whether it passes through the ring.

- ☐ Passed through
- ☐ Not passed through

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